# Bird Species Observation Analysis

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## 1. Executive Summary

This project analyses bird species distribution and diversity across Forest and Grassland habitats using real observational data. Through data cleaning, PostgreSQL analysis, and interactive Power BI dashboards, actionable insights were generated for conservation, eco-tourism, and land management.

## 2. Problem Statement & Objectives

The goal was to understand habitat preferences, identify biodiversity hotspots, seasonal patterns, weather effects, and conservation priorities for bird species.

## 3. Approach & Methodology

- Cleaned and combined multiple Excel sheets using Python.
- Loaded data into PostgreSQL and performed 18+ EDA queries.
- Developed **3 interactive dashboards** in Power BI for dynamic exploration.

## 4. Key Findings

**4.1 Habitat Comparison** Forest: 8,546 observations, 108 unique species, 408 plots
Grassland: 8,531 observations, 107 unique species, 201 plots

**4.2 Top Species** Northern Cardinal (1,160), Carolina Wren (1,011), Red-eyed Vireo (739), etc.

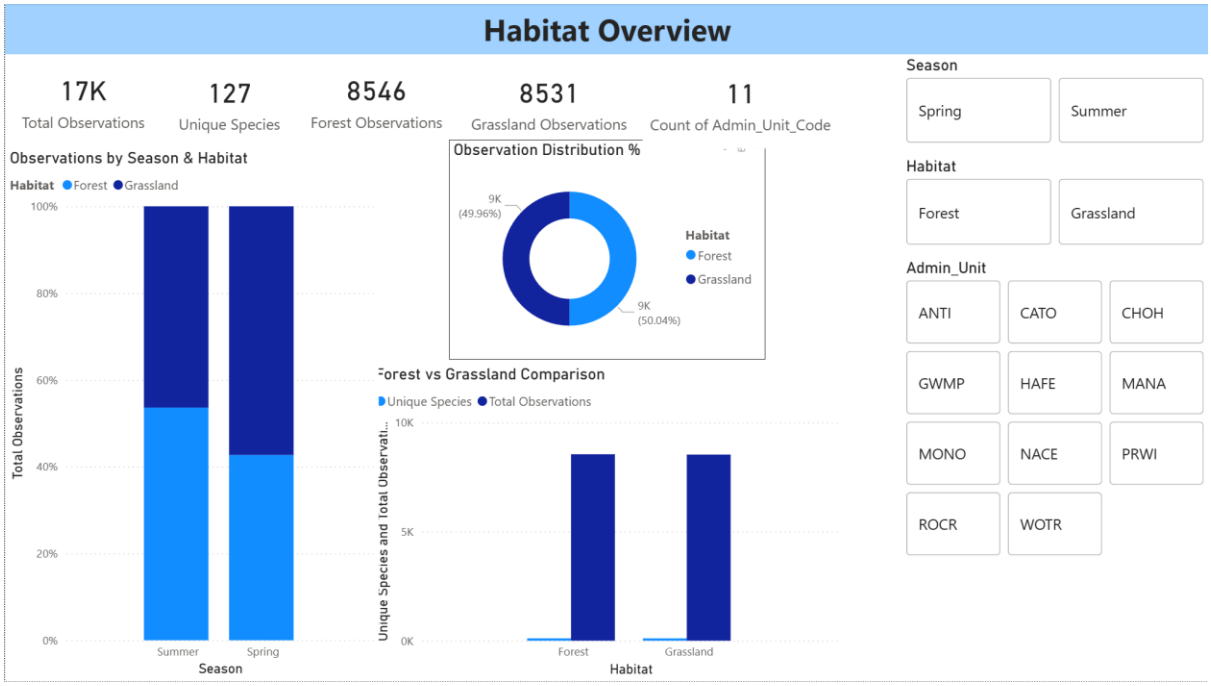
**4.3 Species Habitat Preference** American Goldfinch strongly prefers Grassland (71%), European Starling almost exclusively in Grassland (99%).

**4.4 Biodiversity Hotspots** ANTI (Grassland) and MONO (Grassland) are the top hotspots.

**4.5 Conservation** Wood Thrush is the highest priority at-risk species with 309 observations.

## 5. Interactive Power BI Dashboards

### Dashboard 1: Habitat Overview

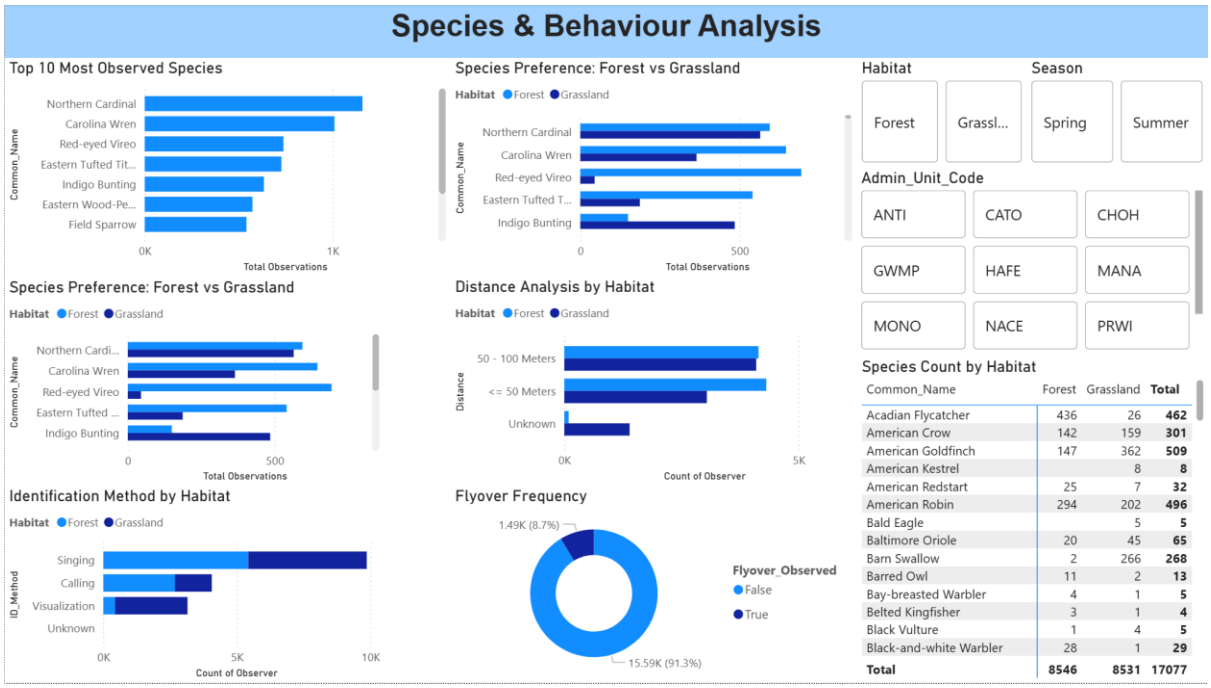



**Explanation:**

This dashboard serves as the executive summary view. It provides key performance indicators such as Total Observations (17,077), Unique Species (127), Forest Observations (8,546), Grassland Observations (8,531), and Count of Admin Units (11).

The stacked bar chart shows seasonal patterns by habitat, while the donut chart illustrates the nearly equal distribution of observations between Forest and Grassland. The bottom clustered bar compares unique species and total observations between the two habitats. This page helps stakeholders quickly understand the overall scale and balance between the two ecosystems. Users can filter by Habitat, Season, and Admin Unit for interactive exploration.

## Dashboard 2: Species & Behaviour Analysis




| Common_Name | Forest | Grassland | Total |
|---|---|---|---|
| Acadian Flycatcher | 436 | 26 | **462** |
| American Crow | 142 | 159 | **301** |
| American Goldfinch | 147 | 362 | **509** |
| American Kestrel | | 8 | **8** |
| American Redstart | 25 | 7 | **32** |
| American Robin | 294 | 202 | **496** |
| Bald Eagle | | 5 | **5** |
| Baltimore Oriole | 20 | 45 | **65** |
| Barn Swallow | 2 | 266 | **268** |
| Barred Owl | 11 | 2 | **13** |
| Bay-breasted Warbler | 4 | 1 | **5** |
| Belted Kingfisher | 3 | 1 | **4** |
| Black Vulture | 1 | 4 | **5** |
| Black-and-white Warbler | 28 | 1 | **29** |
| **Total** | **8546** | **8531** | **17077** |

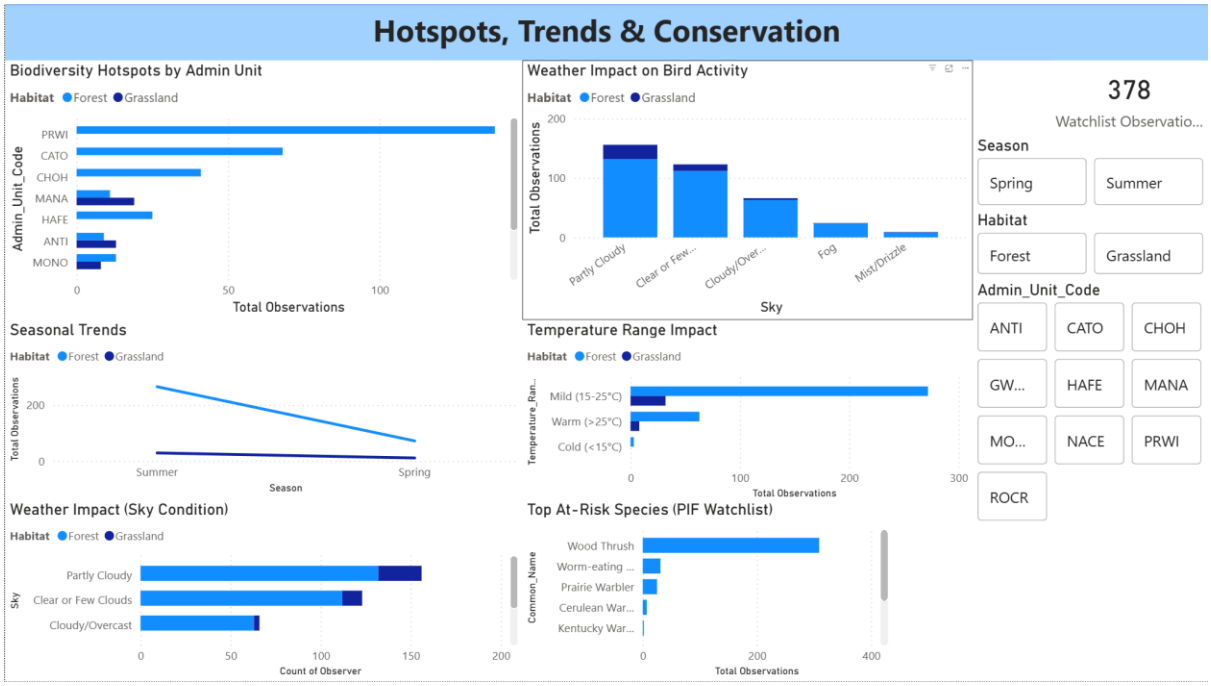
**Explanation:**

This dashboard focuses on detailed species-level and behavioural insights. It includes:

- Top 10 most observed species (Northern Cardinal leads with 1,160 observations).
- Species preference comparison between Forest and Grassland (e.g., American Goldfinch is 71% in Grassland).
- Distance analysis showing most birds are observed within 50–100 meters.
- Identification method breakdown (Singing, Calling, Visualization) by habitat.
- Flyover frequency analysis.

This page is particularly useful for understanding behavioural patterns and species-specific habitat preferences, helping in targeted conservation planning.

## Dashboard 3: Hotspots, Trends & Conservation




**Explanation:**

This dashboard delivers actionable, decision-oriented insights. It features:

- Biodiversity hotspots by Admin Unit (ANTI Grassland leads with 3,588 observations).
- Seasonal trends showing higher activity in Spring and Summer.
- Weather impact analysis (highest observations under Partly Cloudy skies).
- Temperature range impact.
- Top at-risk species from the PIF Watchlist (Wood Thrush dominates with 309 observations).

This page is designed for conservationists and policymakers, providing clear priorities for habitat protection and monitoring efforts.

**Key Features Across All Dashboards:**

- Interactive slicers with cross-filtering
- Consistent colour scheme (Blue tones for Forest, Dark Blue for Grassland)
- Navigation buttons between pages
- Top N filters for focused analysis

**6. Business Insights & Recommendations**

1. **Conservation Priority**: Focus protection on Wood Thrush habitats, primarily in Forest areas.
2. **Eco-Tourism**: Develop birdwatching infrastructure in ANTI, MONO, PRWI, and CHOH, targeting Spring season.
3. **Land Management**: Maintain a balanced mix of forest and grassland to support diverse species.
4. **Monitoring**: Regular surveys in high-activity hotspots (ANTI & MONO) during peak seasons.

**7. Tools & Technologies Used**

- Data Cleaning: Python (Pandas)
- Database & Analysis: PostgreSQL
- Visualization: Power BI (3 interactive dashboards)

**8. Conclusion**

This project successfully demonstrated an end-to-end analytics workflow — from cleaning complex multi-sheet data to building professional interactive dashboards and generating actionable conservation insights. The findings provide clear guidance for biodiversity management and sustainable environmental planning in the region.